

1. Determine the general term (a_n) of these following sequences.

1.1 The sequence 1, 5, 13, 29, 61, ...

(3 points)

1.2 The sequence $\frac{1}{2 \cdot 3}, \frac{4}{3 \cdot 5}, \frac{7}{4 \cdot 7}, \frac{10}{5 \cdot 9}, \dots$

(3 points)

1.3 The sequence $\frac{1}{1}, \frac{3}{5}, \frac{6}{12}, \frac{10}{22}, \frac{15}{35}, \dots$

(4 points)

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2. Let a, b, c be real numbers which $\frac{1}{a}, \frac{1}{2b}, \frac{1}{3c}$ is a geometric sequence

and a, b, c is an arithmetic sequence. Find $\frac{a}{c} + \frac{c}{a}$.

(8 points)

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3.1. Find sum of numbers which are divisible by 3 from 10 to 100.

(5 points)

3.2. In the geometric series, if $a_3 = 36$ and $\sum_{n=1}^3 a_n = 28$. Find a_1 and r .

(5 points)

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4.1. Let $b = 4 + \frac{4}{3} + \frac{4}{9} + \frac{4}{27} + \frac{4}{81} + \dots$. (5 points)

Find sum of the first b terms of $4 \cdot 3 + 7 \cdot 5 + 10 \cdot 7 + 13 \cdot 9 + \dots$. (using $\sum$ notation)

4.2. Find sum of the series $\frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \frac{1}{7 \cdot 9} + \frac{1}{9 \cdot 11} + \frac{1}{11 \cdot 13} + \dots$. (5 points)

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5.1. Given $a_n = (-1)^n (\sqrt{n^2 + 9n} - \sqrt{n^2 - 7})$.

Find $\lim_{n \rightarrow \infty} a_n$ or state that it does not exist and explain your reasoning.

(4 points)

5.2. Given $a_n = \frac{\sqrt{36n} - \sqrt[3]{5n-1} + 11}{5 - 3\sqrt[4]{n} - \sqrt{4n}} + \frac{5^{n+1}}{2^n - 5^n}$.

Find $\lim_{n \rightarrow \infty} a_n$ or state that it does not exist and explain your reasoning.

(4 points)

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6. Given that $A(3,0)$, $B(1,2)$, $C(0,4)$, and $D(2,3)$. Find

6.1. $\overrightarrow{AB}$ and $\overrightarrow{CD}$

(2 points)

6.2. $|\overrightarrow{AB} + 3\overrightarrow{CD}|$

(3 points)

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6.3. the vector that has a length of $|\overrightarrow{AB}|$ and has the opposite direction of $\overrightarrow{CD}$.

(4 points)

7.1. Assume that $\vec{A} = 16\hat{i} + a\hat{j}$ and $\vec{B} = 8\hat{i} + b\hat{j}$ where a and b are the real numbers.

If the angle between $\vec{A}$ and $\vec{B}$ is 60° , find $(a + b)^2$. (5 points)

7.2. Assume that $|\vec{u}| = 5$, $|\vec{v}| = 3$, and $|\vec{u} + \vec{v}| = 4$. Find $|\vec{u} - \vec{v}|$. (5 points)

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8. Given $\vec{u} = 3\hat{i} - \hat{j} + 2\hat{k}$, $\vec{v} = 2\hat{i} - 4\hat{j} + 3\hat{k}$, and $\vec{w} = -2\hat{i} + (3\lambda - 2)\hat{j} + \lambda\hat{k}$.

8.1. Find $\vec{u} \times \vec{v}$.

(3 points)

8.2. Find the value of λ such that $\vec{u}$ and $\vec{w}$ are orthogonal.

(4 points)

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8.3. Assume that $\lambda = 1$. Find the volume of parallelepiped created from $\vec{u}$, $\vec{v}$, and $\vec{w}$.

(4 points)